財政學 HW6

繳交期限：2023年12月20日 23:59

1. 試舉一例說明利率小於等於內部報酬率，但淨現值為負的情形 [提示：比較情境 A 與情境 B。情境 A：第零期只有成本，第一期只有利益；情境 B：第零期只有利益，第一期只有成本]

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情境 A：第零期的成本 x，第一期利益 y

$$-x+\frac{y}{1+IRR}=0\ \rightarrow\ x=\frac{y}{1+IRR}\ \rightarrow 1+IRR=\frac{y}{x}\rightarrow\ IRR=\frac{y}{x}-1$$

$$-x+\frac{y}{1+r}<0\ \rightarrow\ x>\frac{y}{1+r}\ \rightarrow 1+r>\frac{y}{x}\rightarrow\ r>\frac{y}{x}-1$$

$r>IRR$ 不符合題意；

情境 B：第零期的利益 x，第一期成本 y

$$x-\frac{y}{1+IRR}=0\ \rightarrow\ x=\frac{y}{1+IRR}\ \rightarrow 1+IRR=\frac{y}{x}\rightarrow\ IRR=\frac{y}{x}-1$$

$$x-\frac{y}{1+r}<0\ \rightarrow\ x<\frac{y}{1+r}\ \rightarrow 1+r<\frac{y}{x}\rightarrow\ r<\frac{y}{x}-1$$

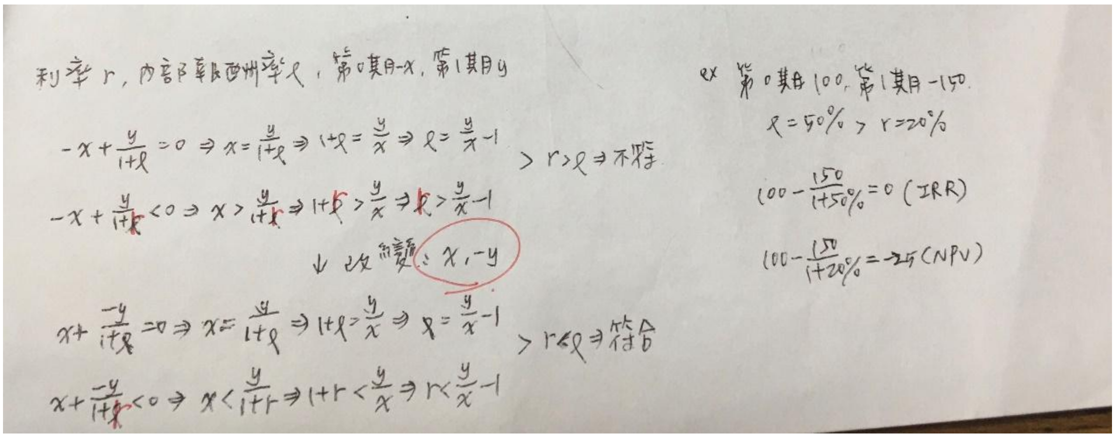
$r<IRR$ 符合題意。

例如第零期的利益 100，第一期成本 150

IRR=50% > r=20%

$$100-\frac{150}{1+50\%}=0\ (\text{算}IRR)$$

$$100-\frac{150}{1+20\%}=-25\ (\text{NPV})$$



利率 r, 內部報酬率 ρ, 第0期 -x, 第1期 y

$-x+\frac{y}{1+\rho}=0 \Rightarrow x=\frac{y}{1+\rho} \Rightarrow 1+\rho=\frac{y}{x} \Rightarrow \rho=\frac{y}{x}-1$ > $r>\rho$ ⇒ 不符

$-x+\frac{y}{1+r}<0 \Rightarrow x>\frac{y}{1+r} \Rightarrow 1+r>\frac{y}{x} \Rightarrow r>\frac{y}{x}-1$

↓ 改編：x, -y

$x+\frac{-y}{1+\rho}=0 \Rightarrow x=\frac{y}{1+\rho} \Rightarrow 1+\rho=\frac{y}{x} \Rightarrow \rho=\frac{y}{x}-1$ > $r<\rho$ ⇒ 符合

$x+\frac{-y}{1+r}<0 \Rightarrow x<\frac{y}{1+r} \Rightarrow 1+r<\frac{y}{x} \Rightarrow r<\frac{y}{x}-1$

ex 第0期 100, 第1期 -150

$\rho=50\%>r=20\%$

$100-\frac{150}{1+50\%}=0$ (IRR)

$100-\frac{150}{1+20\%}=-25$ (NPV)

2. Conduct a project evaluation with all three cost-benefit analysis methods using the information provided in the last two slides (Hint: 用 Excel 計算上課投影片最後兩頁的資訊)

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(1)淨現值法則：兩個計劃都不可行，因為淨現值都<0：-$4434、-$3060

(2)內部報酬率法則：內部報酬率約為 0.94%，皆<題目給定的折現率，因此都不可行

(3) 益本比法則：兩個計劃都不可行，因為益本比都<1：1379/5813=0.237226905、4060/7120=0.570224719

3. (Rosen 8.3) A project yields an annual benefit of $25 a year, starting next year and continuing forever. What is the present value of the benefits if the interest rate is 10 percent? [Hint: 由等比級數公式推導 The infinite sum x+x2+x3+… is equal to x/(1 - x), where x is a number less than 1.] Generalize your answer to show that if the perpetual annual benefit is B and the interest rate is r, then the present value is B/r.

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The present value of $25/.10 = $250.

The present value of the perpetual annual benefit = $B/(1 + r) + B/(1 + r)^2 + \ldots = B (r + 1 - r)/r = B/r$.

4. (Rosen 8.4) Suppose that you are planning to take a year vacation to bike across the United States. Someone is willing to sell you a new bicycle for $500. At the end of the year, you expect to resell the bicycle for $350. The benefit to you of using the bicycle is the equivalent of $170.

(a) What is the internal rate of return?

(b) If the discount rate is 5 percent, should you buy the bicycle?

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a. The internal rate of return is the discount rate that would make the project's net present value (NPV) equal zero. To solve for the internal rate of return, ρ, set the present value of benefits minus the present value of costs equal to zero. If we assume the benefit of using the bicycle is immediate (and worth $170), there is also the benefit of re-selling the bicycle for $350, but it can't be re-sold until next year, so must be discounted. Therefore, NPV is 170 + [350/(1+ρ)] – 500 = 0. Solving this expression for ρ yields ρ = 6 percent. If we assume that the benefits of the vacation will not be enjoyed for one year, then NPV is [(170+350)/(1+ρ)] –500 and setting this expression equal to zero and solving for ρ yields ρ = 4 percent.

b. If the discount rate is 5 percent, purchasing the bicycle is a good idea if ρ is 6 percent, but a bad idea if ρ is 4 percent.

5. (Hyman 6.2) Suppose a proposed new road to be constructed in North Carolina between Raleigh and Morehead City will lower the average cost per trip by car from $5 to $4. Currently, 500,000 trips are made between the two cities per year. An estimate indicates that, all other things being equal, the new road will increase the number of trips per year to 600,000. Calculate the annual benefits to motorists of the new road as based on their willingness to pay.

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There is a $1 per trip cost savings on the existing 500,000, which provides a benefit of $500,000 per year. The gain in net benefits from the 100,000 new trips is 1/2($1)(100,000) = $50,000 per year. The annual increase in benefits to motorists as a result of the new road will therefore be $550,000.